# 3.1.2 Amount of substance: Sheet 11

*Percentage yield and the ideal gas equation · 17 marks · about 25 minutes*

**Key information**

- **pV = nRT**: p in Pa, V in $m^3$, T in K, R = 8.31 J $K^{-1}$ $mol^{-1}$
- Unit conversions: kPa × 1000 → Pa; $cm^3 \times 10^{-6} \rightarrow m^3$; °C + 273 → K.
- Limiting reagent: compare the moles of each reactant using the equation ratio; the one that runs out first limits the product.
- **% yield = actual mass ÷ theoretical mass × 100**; **% atom economy = $M_r$ of desired product ÷ total $M_r$ of reactants × 100**
- Mass of a liquid = volume × density.

## Section A: Self-quiz

**01** This is the equation for the conversion of 1,2-dibromoethane to butanedinitrile.

$CH_2BrCH_2Br + 2\ KCN \rightarrow NCCH_2CH_2CN + 2\ KBr$

Relative formula masses: $CH_2BrCH_2Br$ = 187.8, KCN = 65.1, $NCCH_2CH_2CN$ = 80.0, KBr = 119.0

What is the percentage atom economy for the production of butanedinitrile?

**[1 mark]**

**A** 100%
**B** 40.2%
**C** 31.6%
**D** 25.2%

**02** When heated, a sample of potassium chlorate(V) ($KClO_3$) produced 67.2 $cm^3$ of oxygen, measured at 298 K and 110 kPa

$2\ KClO_3(s) \rightarrow 2\ KCl(s) + 3\ O_2(g)$

What is the amount, in moles, of potassium chlorate(V) that has decomposed? The gas constant, R = 8.31 J $K^{-1}$ $mol^{-1}$

**[1 mark]**

**A** $9.95 \times 10^{-4}$
**B** $1.99 \times 10^{-3}$
**C** $2.99 \times 10^{-3}$
**D** $4.48 \times 10^{-3}$

**03** 130 $cm^3$ of oxygen and 40 $cm^3$ of nitrogen, each at 298 K and 100 kPa, were placed into an evacuated flask of volume 0.50 $dm^3$. What is the pressure of the gas mixture in the flask at 298 K?

**[1 mark]**

**A** 294 kPa
**B** 68.0 kPa
**C** 34.0 kPa
**D** 13.7 kPa

**04** What is the minimum volume, in $dm^3$, of air needed for the complete combustion of 1 $dm^3$ of methane? Assume that air contains 20% of oxygen by volume and that all volumes are measured at the same temperature and pressure.

**[1 mark]**

**A** 1
**B** 2
**C** 5
**D** 10

**05** The equation below represents the complete combustion of butane.

$C_4H_{10}(g) + 6½O_2(g) \rightarrow 4CO_2(g) + 5H_2O(g)$

20 $cm^3$ of butane are completely burned in 0.20 $dm^3$ of oxygen. Which statement is correct? All volumes are measured at the same temperature and pressure.

**[1 mark]**

**A** 40 $cm^3$ of carbon dioxide are formed
**B** 0.065 $dm^3$ of oxygen react
**C** 70 $cm^3$ of oxygen remain
**D** 0.50 $dm^3$ of steam are formed

## Section B: Exam questions

**06** An alkene ($M_r$ = 98.0) is made by heating an alcohol ($M_r$ = 116.0) with concentrated phosphoric acid. One mole of alcohol forms one mole of alkene.

In an experiment, 12.0 $cm^3$ of the alcohol produces 6.12 g of the alkene.

Density of the alcohol = 0.818 g $cm^{-3}$

Calculate the percentage yield.

**[5 marks]**

**07** Magnesium reacts with hydrochloric acid.

$Mg(s) + 2\ HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

0.400 g of magnesium is added to 20.0 $cm^3$ of 1.50 mol $dm^{-3}$ hydrochloric acid.

Identify the limiting reagent.

Justify your answer.

Calculate the volume, in $m^3$, of hydrogen produced at 101 kPa and 15 °C

The gas constant, R = 8.31 J $K^{-1}$ $mol^{-1}$

**[7 marks]**

# Mark Scheme

*M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.*

| Question | Marking guidance | Comments | Mark |
|---|---|---|---|
| 01 | **D** | Source: 7404/2 Jun 2024 Q21 | 1 |
| 02 | **B** | Source: 7405/3 Nov 2020 Q07 | 1 |
| 03 | **C** | Source: 7405/3 Jun 2017 Q34 | 1 |
| 04 | **D** | Source: 7404/2 Jun 2023 Q17 | 1 |
| 05 | **C** | Source: 7404/1 Jun 2019 Q21 | 1 |
| 06 | M1 mass of alcohol = 12.0 × 0.818 (= 9.82 g)<br>M2 amount of alcohol = M1 ÷ 116(.0) (= 0.0846 mol)<br>M3 $M_r$ of alkene = 98(.0)<br>M4 mass of alkene expected = M2 × M3 (= 8.29 g)<br>M5 % yield = 6.12 ÷ M4 × 100 = 73.8% (at least 2 sf) | Source: 7404/2 Jun 2023 Q07.3 (compound names replaced; $M_r$ of alkene and 1 : 1 ratio stated)<br>Allow ECF at each stage<br>M5 should be an attempt at their mass (or moles) of alkene achieved ÷ their mass (or moles) expected × 100 | 5 |
| 07 | M1 moles Mg = 0.400 ÷ 24.3 = 0.0165 mol<br>M2 moles HCl = 20.0 × 1.50 ÷ 1000 = 0.0300 mol<br>M3 HCl is limiting, with justification (eg 0.0165 mol of Mg requires 0.0330 mol of HCl; or moles of HCl is less than double the moles of Mg)<br>M4 amount of $H_2$ formed = 0.0150 (or M2 ÷ 2)<br>M5 converting T to 288 K, p to 101 000 Pa<br>M6 V = (M4 × 8.31 × 288) ÷ 101 000<br>M7 V = $3.55 \times 10^{-4}$ ($m^3$) | Source: 7404/2 Jun 2024 Q05.1<br>Allow ECF at each stage; M4 independent of M3<br>Answer based on 0.0165 mol Mg as limiting reagent gives 3.79 – 3.91 × $10^{-4}$ $m^3$ which scores M5, M6, M7 and possibly M3, M4<br>If 1 mol = 24 $dm^3$ used rather than the ideal gas equation, cannot score M5/6/7 | 7 |